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Dynamic Quantum-Light Radiation Shaping with Time-Modulated BTO-Driven Nanoantennas Array

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Abstract

Quantum nanophotonics enables advanced functionalities in information processing and sensing by providing control over quantum-light emission. Introducing time modulation into nanophotonic environments adds an additional dimension, enabling dynamic reconfiguration of the local density of optical states (LDOS), access to Floquet harmonics, and simultaneous control of emission rate, spectrum, and direction capabilities that are fundamentally inaccessible in static platforms. Designing such systems requires self-consistent treatment of temporal modulation, electrodynamics, and quantum dynamics. Here, we extend a quantum–electromagnetic framework by coupling time-modulated Maxwell electrodynamics, solved using an anisotropic discrete dipole approximation, with a harmonic-resolved Gorini–Kossakowski–Sudarshan–Lindblad master equation. This Floquet–Lindblad–DDA approach captures the quantum feedback induced by temporal modulation and enables evaluation of harmonic-resolved populations, radiated and dissipated power, Purcell enhancement, and quantum efficiency. We applied the method to control the emission of a diamond NV center using voltage-driven Pockels modulation of a barium titanate nanoantenna array. The optimized structure achieves Purcell enhancements exceeding 5000, routes more than 75% of the power to the $n = +1$ Floquet harmonic, enabling electronic beam steering by more than 50°. These results demonstrate a platform for programmable quantum-light emission based on time-modulated nanophotonics.

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Introduction

Quantum technologies promise a paradigm shift in information processing, communication, and sensing, where non-classical principles enable functionalities that transcend the limits of conventional photonic platforms [1, 2]. Quantum emitters (QEs) constitute the fundamental building blocks of these systems, as they generate quantum light for applications spanning quantum communication [3], secure free-space optical links, single-photon sources [4, 5], quantum networking, and high-precision metrology. Their integration into photonic architectures underpins reliable quantum computing via spin–photon interfaces and enables enhanced sensing performance based on squeezed and entangled states. Achieving these capabilities requires strong, controllable light–matter interaction, enabling precise control of emission rate, spectrum, polarization, radiation pattern, and detections [5, 6, 1, 7, 8]. Classical and nanophotonic structures—including plasmonic nanoantennas, Mie resonators, waveguides, and photonic crystal cavities—are routinely employed to engineer the local photonic density of states (PDoS), thereby boosting spontaneous emission via resonant enhancement mechanisms [9, 10, 11]. According to Fermi's golden rule, increasing the PDoS increases the number of accessible optical modes, thereby increasing the decay rate Γ. The ratio Γ/Γ_0, known as the Purcell factor [12, 13], is therefore an important figure of merit for modulating QE lifetime, radiative efficiency, and modulation bandwidth.

A widely adopted approach for manipulating the PDoS relies on positioning QEs near strongly resonant or highly dispersive photonic environments such as photonic crystal cavities [14], dielectric nanoholes [10], optical nanoantennas, or hyperbolic metamaterials (HMMs) [9, 13]. These structures reshape the electromagnetic modal fields in their vicinity, enabling both enhancement and suppression of spontaneous emission. Nonetheless, once fabricated, the system is static, and radiation manipulation can only be achieved by changing QE's excitation parameters. In this context, time-modulated photonic structures emerge as powerful platforms for manipulating different classes of quantum emitters, ranging from excitonic emitters in two-dimensional transition metal dichalcogenides (TMDCs) [15, 16] to solid-state defect centers such as diamond nitrogen-vacancy (NV) systems [17, 18], by dynamically engineering the local density of optical states (LDOS) [8]. By tailoring the geometry and material composition of their constituent meta-atoms, these systems enable essential quantum functionalities, including directional routing of single-photon emission, polarization control mediated by surface plasmon polaritons (SPPs) [19], enhanced chiral light–matter interactions in stacked quantum metasurfaces, and the generation of complex entangled states through resonant nonlocal and nonlinear interactions [4, 3].

Beyond the static limit, Floquet-driven and time-modulated photonic structures enable dynamic manipulation of both classical and quantum light, supporting functionalities such as active beam steering, dynamic tuning of the Purcell factor, on-demand spectral shaping of emission, and reconfigurable wavefront and polarization control [20, 21, 22, 23, 24]. Furthermore, they provide precise control over radiation patterns, polarization, and wavefronts by tailoring their band structure, local density of states, and spatial dispersion, enabling a plethora of applications across efficient frequency conversion through Floquet sideband engineering [25, 26, 27, 28, 11]. Spatially periodic media, including photonic crystals and metasurfaces, further provide precise control over radiation patterns, polarization, and wavefronts by tailoring their band structure, local density of states, and spatial dispersion, enabling a plethora of applications across various fields [9, 29, 30, 31].

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In this context, the emergence of diverse material platforms, including transparent conducting oxides such as indium tin oxide (ITO), excitonic van der Waals materials such as transition-metal dichalcogenides (TMDCs) and aligned carbon nanotubes (CNTs), as well as nonlinear crystals such as lithium niobate has opened promising avenues for realizations of active and Floquet-driven metasurfaces [16, 18, 32, 33, 34, 35, 36]. Notably, space–time metasurfaces based on ITO have been experimentally demonstrated to generate tailored optical sidebands and simultaneously manipulate their spatial content at optical frequencies, thereby establishing a viable route toward dynamic spectral–spatial control in time-modulated photonic platforms [37, 38, 39]. Among electro-optic materials, barium titanate (BTO) has recently emerged as an exceptional platform due to a combination of a large Pockels coefficient, broad optical transparency, and compatibility with integrated photonic architectures [40, 41]. Recent measurements in both thin-film and waveguide configurations have confirmed that BTO can exhibit Pockels coefficients exceeding 600 pm/V, approaching and even matching bulk values [41]. This extraordinarily strong electro-optic response enables high-speed, low-power nanoscale refractive-index modulation, establishing BTO as a particularly attractive candidate for implementing time-varying devices for the dynamic control of both quantum and classical light.

The electromagnetic response of these time-modulated photonic platforms can be addressed in different ways. For periodically patterned structures, analytical and semi-analytical descriptions can be obtained using space-time rigorous coupled-wave analysis (RCWA), which provides full-wave solutions for temporally modulated periodic media [42]. At the single-particle level, time-Floquet Mie theory enables closed-form analytical insight into parametric resonances, sideband generation, and momentum-bandgap effects in temporally modulated scatterers [43, 44].
For more complex and aperiodic geometries, however, the time-modulated discrete dipole approximation (TM-DDA) becomes a particularly powerful and general framework [45, 43]. Unlike RCWA and time-Floquet Mie theory, DDA allows modeling non-spherical and non-periodic structures, which is fundamental for more practical applications. In addition, material platforms such as barium titanate (BTO) naturally exhibit strong anisotropy and off-diagonal electro-optic components under applied bias. In such cases, the local dielectric response must be explicitly diagonalized at each dipole site to correctly construct the polarizability tensor used in the DDA formulation [46, 47]. Although diagonalization has been performed, its application to time-modulated materials remains to be realized. In this sense, we extend current formulations of TM-DDA to account for off-diagonal elements of the permittivity tensor, threby enabling the simulation of novel photonics structures with anisotropic materials, such as BTO. Although TM-DDA can solve the electromagnetic part of the quantum problem, a fully self-consistent description of quantum emitters embedded in time-modulated photonic platforms remains a fundamentally challenging task, as it requires bridging temporally varying classical electrodynamics with open-quantum-system dynamics. While several works approximate the QE as a classical dipole with a fixed polarizability [48, 49, 50, 9], this approach neglects essential quantum features such as population transfer, coherence buildup, and feedback between the emitter and its photonic environment.

More advanced hybrid schemes have attempted to couple GKSL-type quantum dynamics with static electromagnetic solvers [48, 51, 52, 53], but these methods do not explicitly account for the temporal modulation's influence on the electromagnetic modes surrounding the emitter. Open quantum systems are commonly modeled through Marko-

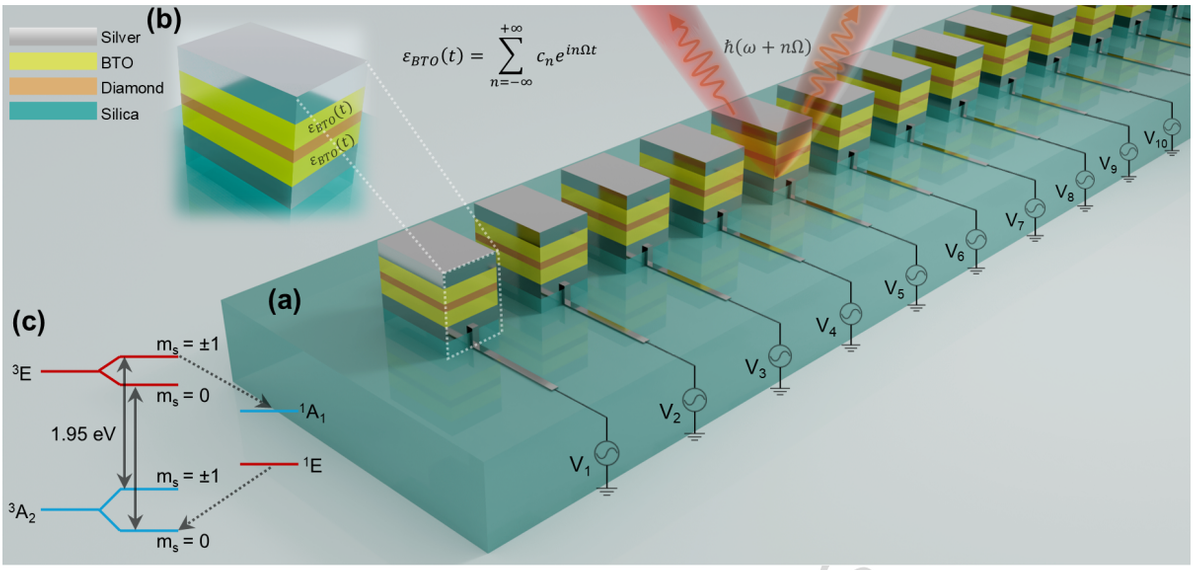


Figure 1: **Floquet-driven BTO meta-atoms for dynamic control of quantum emission.** (a) Schematic of the one-dimensional array of BTO-based meta-atoms, each driven by an independently modulated voltage to dynamically control the radiation of a quantum emitter hosted in a diamond NV center. (b) Structure of a single MIM cavity, highlighting its constituent materials and the time-varying permittivity of the BTO layer. (c) Energy level diagram of the diamond NV center used as the quantum emitter in this work.

vian master equations, most prominently the Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) formalism, which provides a physically consistent description of dissipative quantum dynamics by incorporating spontaneous emission, dephasing, and incoherent pumping through Lindblad superoperators [54, 55, 56].

When the system Hamiltonian and the dissipative channels become explicitly time-periodic, the emitter dynamics can be described within the Floquet–GKSL formalism, where the master equation is formulated in an extended Sambe space that captures harmonic coupling between Floquet sidebands of the driven system. Such formulations have been rigorously developed for periodically driven open quantum systems, in which temporal modulation induces a harmonic structure in both coherent and dissipative channels [57, 58, 29]. However, in existing implementations, the electromagnetic environment is typically treated as either static, weakly perturbed, or analytically prescribed, and the harmonic structure of the decay channels is not extracted from a full-wave, time-modulated electromagnetic simulation of realistic nanophotonic geometries. This framework has proven valuable for analyzing light–matter interactions in complex photonic environments, including plasmonic and hybrid nanostructures, where fully quantum-optical treatments reveal rich phenomena such as emitter–plasmon hybridization, polaritonic mode formation, and long-lived quantum coherences [59, 60, 61].

In this sense, we develop a time-modulated quantum metasurface platform based on BTO-assisted, Floquet-driven time-varying nanoantenna arrays, and extend a hybrid modeling approach that couples the Floquet–GKSL formalism with a time-modulated DDA solver. This approach enables active, reconfigurable control of quantum emission through space-time engineering of the electromagnetic environment, as schematically illustrated in Fig. 1(a,b). The system consists of a one-dimensional array of metal–

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insulator–metal (MIM) nanoantennas loaded with a BTO electro-optic layer, where each meta-atom is independently driven by a time-dependent voltage $V_i(t)$. This configuration enables programmable spatiotemporal control of the local density of optical states (LDOS), allowing dynamic manipulation of the emission properties of an embedded quantum emitter.

As the quantum light source, we consider a diamond nitrogen-vacancy (NV) center, whose zero-phonon line (ZPL) is centered at 637 nm, making it fully compatible with the designed optical response of the nanoantenna array (Fig. 1(c)). By exploiting the strong Pockels effect in BTO, the refractive index of each meta-atom can be modulated at radio-frequency or microwave rates, thereby inducing Floquet sidebands in the system's electromagnetic response. This enables simultaneous control of the spontaneous emission rate, the redistribution of emitted power among Floquet harmonics, and the selective enhancement or suppression of specific sidebands. Within the proposed approach, the quantum-state evolution of the NV center is directly coupled to the temporally modulated electromagnetic Green's function of the nanoantenna array, resulting from the self-consistent integration of quantum open-system dynamics with a full-wave electromagnetic solver that treats time-modulated, anisotropic meta-atoms arranged in programmable array configurations.

Using this formalism, we demonstrate active beam steering of quantum emission enabled by time modulation within a self-consistent classical–quantum solver. In particular, the radiated power can be preferentially routed into a selected Floquet harmonic, enabling harmonic-selective quantum emission. Moreover, by independently controlling the modulation phase of each voltage channel $V_i(t)$ (Fig. 1(c)), the radiation phase profile across the array can be dynamically engineered. This enables real-time beam steering of selected Floquet harmonics without any mechanical motion, providing spatiotemporal control over the emitted quantum light. Such a capability enables programmable quantum radiation patterns, directional single-photon sources, and dynamically reconfigurable quantum photonic circuits. Altogether, the proposed BTO-assisted Floquet-driven nanoantenna array establishes a practical framework for for space-time programmable quantum photonics.

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Results

To enable a self-consistent description of quantum emission in time-modulated nanoantenna arrays, we extend the modeling framework by combining a quantum open-system model and the DDA solver for single and for an array of nanoantennas, as depicted in Fig. 2 (a), (b), and (c). On the classical side, the nanophotonic structure is discretized into a lattice of anisotropic voxels using a time-modulated discrete dipole approximation (DDA). On the quantum side, the emitter dynamics are described by the Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation. The two parts are coupled self-consistently through the harmonic electric field and the harmonic dipole moments, enabling a step-by-step time evolution of the harmonic density operator and its feedback onto the electromagnetic response. After establishing the single-emitter formulation, the approach is naturally extended to the nanoantenna array geometry through homogenization and inter-particle coupling.

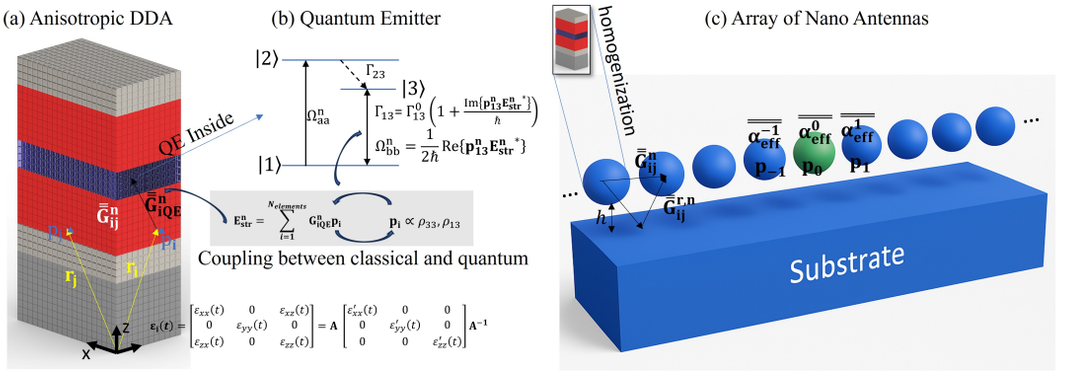


Figure 2: (a) Anisotropic time-modulated DDA representation of the multilayer unit cell. The structure is discretized into anisotropic voxels whose permittivity varies over time, resulting in harmonic dipole responses. (b) Three-level quantum emitter model and its coupling to the classical fields. The time-modulated fields modify both the Rabi frequencies and the decay rates, enabling harmonic-resolved spontaneous and stimulated emission. The quantum dipole moments $\mathbf{d}_{31}^{n}$ feed back into the classical DDA as external sources. (c) Homogenized description of the modulated multilayer and its use in extended arrays of nanoantennas. The full harmonic polarizability tensor $\overline{\overline{\boldsymbol{\alpha}}}_{\text{eff}}^{n,k}$ is extracted by illuminating the homogenized unit cell with individual harmonic components of the incident field.

GKSL equation

For the quantum part of the proposed modeling, we consider an N-level quantum system with eigenstates $|i\rangle$ and energies $\hbar\omega_i$. The emitter dynamics are described using the Markovian master equation in its standard Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) form [54, 56, 62, 29, 63]:

$$\frac{d\hat{\rho}(t)}{dt} = -\frac{i}{\hbar}\left[\hat{H}(t), \hat{\rho}(t)\right] + \sum_k \Gamma_k \left(\hat{L}_k \hat{\rho}(t) \hat{L}_k^\dagger - \frac{1}{2}\left\{\hat{L}_k^\dagger \hat{L}_k,\, \hat{\rho}(t)\right\}\right), \tag{1}$$

where $\hat{L}_k$ are the Lindblad jump operators and Γ_k are the corresponding decay rates, $\hat{\rho}$ is the density operator, $\hat{H}$ is the system Hamiltonian. Under the dipolar approximation, the Hamiltonian $\hat{H}$ can be written as

$$\hat{H}(t) = \sum_i \hbar\omega_i |i\rangle\langle i| - \hat{\mathbf{d}} \cdot \mathbf{E}_{\text{loc}}(\mathbf{r}, t) \tag{2}$$

where $\mathbf{E}_{\text{loc}}(\mathbf{r}, t)$ is the harmonic-resolved local electric field obtained from the time-modulated DDA solver and $\hat{\mathbf{d}}$ is the dipole operator. Because both the Hamiltonian and the electromagnetic environment are explicitly time-periodic with modulation frequency ω_m, the system is treated within a Floquet framework. Applying this expansion to the density operator,

$$\hat{\rho}(t) = \sum_n \hat{\rho}^{\,n} e^{in\omega_m t}, \tag{3}$$

and substituting into Eq. (1) yields a coupled set of equations among the different harmonic components $\hat{\rho}^n$. This can be written compactly in Liouville (vectorized) form

as

$$\frac{d}{dt}\begin{bmatrix}\mathrm{vec}(\hat{\rho}^{-N_h})\\ \vdots \\ \mathrm{vec}(\hat{\rho}^{0})\\ \vdots \\ \mathrm{vec}(\hat{\rho}^{+N_h})\end{bmatrix} = \mathbf{L}(t)\begin{bmatrix}\mathrm{vec}(\hat{\rho}^{-N_h})\\ \vdots \\ \mathrm{vec}(\hat{\rho}^{0})\\ \vdots \\ \mathrm{vec}(\hat{\rho}^{+N_h})\end{bmatrix}, \tag{4}$$

where $\mathbf{L}(t)$ is the Floquet Liouvillian superoperator. The matrix $\mathbf{L}(t)$ contains diagonal blocks corresponding to the GKSL Liouvillian of each harmonic and off-diagonal blocks that couple harmonics through the time-modulated field $\mathbf{E}^n_{\mathrm{loc}}$ obtained from the DDA solver. This structure explicitly captures the harmonic mixing induced by temporal modulation. At each time step t_k, the local field harmonics define the instantaneous Rabi frequencies and decay rates, which determine the Liouvillian superoperator $\mathbf{L}(t_k)$ associated with Eq. (1). The density operator is propagated through exponential integration of the Liouvillian:

$$\mathrm{vec}[\hat{\rho}(t_{k+1})] = \exp[\mathbf{L}(t_k)\,\Delta t]\,\mathrm{vec}[\hat{\rho}(t_k)]\,, \tag{5}$$

where $\mathrm{vec}(\cdot)$ denotes vectorization of the density matrix. This propagation and formalism preserves trace and complete positivity intrinsically. The harmonic components of the density matrix, $\rho^n_{ij}(t)$, are then extracted from the Floquet expansion of $\hat{\rho}(t)$. The full description of the Liouvillian superoperator is shown in the Supplementary Section S2.

When considering the 3-level system, such as the NV diamond adopted here (Fig. 2 (c)), the instantaneous emission rate for the $3\to1$ transition includes both spontaneous and stimulated contributions. The spontaneous emission is proportional to the excited-state population, $R^{3\to1,n}_{\mathrm{sp}} = \Gamma^n_{31}\,\rho^n_{33}$, where ρ^n_{33} denotes the nth Floquet harmonic of the excited-state population obtained from the harmonic expansion of the density matrix, $\rho_{33}(t) = \sum_n \rho^n_{33} e^{in\omega_m t}$.
The stimulated emission is driven by field–dipole coherence, $R^n_{\mathrm{stim}} = 2\,\mathrm{Im}[\omega^n_{31}\,\rho^{n*}_{31}]$, where ρ^n_{31} is the nth harmonic of the optical coherence between states $|3\rangle$ and $|1\rangle$, defined by $\rho_{31}(t) = \sum_n \rho^n_{31} e^{in\omega_m t}$, and ω^n_{31} represents the nth harmonic of the Rabi frequency associated with the $3\leftrightarrow1$ transition, obtained from the harmonic decomposition of the time-modulated local electric field. The total emission rate is obtained by summing both contributions, $R^n_{\mathrm{tot}} = R^n_{\mathrm{sp}} + R^n_{\mathrm{stim}}$. For each Floquet harmonic n, the corresponding emitted power is $P^n_{\mathrm{emit}} = \hbar\,\omega^n_{31}\,R^n_{\mathrm{tot}}$, where $\omega^n_{31} = \omega_{31} + n\,\omega_m$. Finally, the effective transition dipole moment for each harmonic component is determined from the total emitted power as

$$d^n_{\mathrm{eff}} = \sqrt{\frac{12\pi\,\varepsilon_0\,c^3\,P^n_{\mathrm{emit}}}{\left(\omega^n_{31}\right)^4}}, \tag{6}$$

returning the harmonic-resolved dipole moments

$$\mathbf{p}^n_{\mathrm{eff}} = d^n_{\mathrm{eff}}\,\boldsymbol{\varepsilon}_n. \tag{7}$$

The total QE dissipated power for each harmonic is then computed as

$$P^n_{\mathrm{diss}}(t_{k+1}) = \frac{\omega^n_{31}}{2}\,\mathrm{Im}[\mathbf{p}^n_{\mathrm{eff}}(t_{k+1})\cdot\mathbf{E}^{n*}(\mathbf{r}_{\mathrm{QE}},t_k)]\,, \tag{8}$$

which represents the rate at which the emitter transfers energy to the electromagnetic field in the presence of the temporally modulated photonic structure. The harmonic-resolved decay rates are determined by the electromagnetic environment through the local

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density of optical states (LDOS). Using the intrinsic transition dipole $\mathbf{d}_{31}$, the decay-rate modification is written as

$$\Gamma_{31}^{n} = \Gamma_0 \left[\delta_{n0} + \frac{2}{\hbar} \operatorname{Im} \left(\mathbf{d}_{31} \cdot \mathbf{E}_{\mathrm{loc}}^{n} \right) \right], \tag{9}$$

where $\mathbf{E}_{\mathrm{loc}}^{n}$ is the local harmonic electric field generated by the nanoantenna array in response to the intrinsic dipole source (see Supplementary section S4.5).

Importantly, the quantum–classical coupling is implemented in a self-consistent but hierarchical manner. We first solve the quantum-emitter dynamics using the GKSL formalism. At each time step t_k, the GKSL solver updates the density matrix, from which the populations and optical coherences (e.g., ρ_{31}^{n}) are obtained. These quantities determine the emission dynamics of the system, while the intrinsic transition dipole moment $\mathbf{d}_{31}$ is kept fixed and defined by the free-space decay rate. The GKSL evolution therefore governs how the emitter state evolves in time under the influence of a given photonic environment. The electromagnetic environment is computed independently using the time-modulated DDA solver. In this step, the structure is excited by a fixed dipole source with moment $\mathbf{d}_{31}$, which serves as a reference to extract the local Green tensor and the harmonic electric fields at the emitter position. This procedure provides a normalized characterization of the photonic environment, from which the local density of optical states (LDOS) and the harmonic-resolved Purcell factors are obtained. These quantities determine the decay rates Γ_{31}^{n}, which depend only on the geometry, material properties, and modulation profile of the structure, and are therefore fixed for a given configuration. Once the decay rates and local fields are computed, they are used as inputs to the GKSL solver, which propagates the density matrix in time. The emitted and dissipated powers are then evaluated from the resulting dipole amplitudes as a post-processing step. In this framework, the electromagnetic solver defines the environment via the LDOS, while the GKSL dynamics determine the emitter's quantum state, ensuring a consistent separation between intrinsic dipole properties and environmental modifications.

This formulation allows the evaluation of both spontaneous and stimulated emission contributions, yielding harmonic-resolved dipole moments and the total dissipated power of the quantum emitter. Since the modified decay rates and dipole moments depend explicitly on the local field generated by the photonic environment, a fully consistent description of the emission requires coupling the GKSL equation to the classical Maxwell equations. This coupling is implemented through the time-modulated DDA framework, which is discussed next.

Time Modulated DDA

For the classical model, the electromagnetic response of the system is computed using the time-modulated discrete dipole approximation (DDA). Within this framework, the goal is to determine the induced dipole moment of the j-th lattice element, $\mathbf{p}_j^n$, associated with the n-th frequency harmonic, as shown in Fig. 2 (a). The DDA model is coupled to the GKSL equation, since it determines the dipole inside the structure, which acts as an illumination source, as seen in Fig. 2 (b). However, the source of the GKSL equation is also modulated by the field reflected from the structure back to the QE, meaning that both systems should be solved together for a consistent solution.

At each position $\mathbf{r}_j$, the harmonic dipole moment is obtained by multiplying the local total electric field by the effective polarizability tensor, $\mathbf{p}_j^n = \sum_m \overline{\overline{\boldsymbol{\alpha}}}_j^{n,m} \mathbf{E}_{\mathrm{tot}}^m(\mathbf{r}_j)$, where

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$\overline{\overline{\boldsymbol{\alpha}}}_j^{n,m}$ is the polarizability tensor coupling of the m-th harmonic with the n-th frequency of the j-th element. The total field $\mathbf{E}_{\text{tot}}^n(\mathbf{r}_j)$ is composed of the external incident field $\mathbf{E}_{\text{inc}}^n(\mathbf{r}_j)$ and the collective contribution from all N_{elements} dipoles radiating at the same frequency. Note that the incident field is not restricted to a plane-wave excitation; throughout this manuscript, it corresponds to the electromagnetic field radiated by the quantum emitter modeled as a dipole source, which provides the excitation in the DDA framework. Additionally, the temporal modulation of the polarizability induces coupling among different harmonic orders, which must also be accounted for. The resulting equation for the j-th element is written as

$$\mathbf{p}_j^n = \sum_{m=1}^{N_h} \left[\overline{\overline{\boldsymbol{\alpha}}}_j^{n,m} \mathbf{A}_j \mathbf{E}_{\text{inc}}^m + \overline{\overline{\boldsymbol{\alpha}}}_j^{n,m} \mathbf{A}_j \sum_{\substack{i=1 \\ i\neq j}}^{N_{\text{elements}}} \overline{\overline{\mathbf{G}}}_{ij}^m \mathbf{A}_i^{-1} \mathbf{p}_i^m \right]. \tag{10}$$

Here, $\mathbf{A}_i$ is the rotation matrix that diagonalizes the time-dependent permittivity of element i, defining its principal basis. In this basis, the harmonic polarizability is diagonal, $\overline{\overline{\boldsymbol{\alpha}}}_{i,\text{diag}}^q = \text{diag}(\alpha_{xx}^{i,q}, \alpha_{yy}^{i,q}, \alpha_{zz}^{i,q})$, and the corresponding tensor in the global frame is $\overline{\overline{\boldsymbol{\alpha}}}_i^q = \mathbf{A}_i \overline{\overline{\boldsymbol{\alpha}}}_{i,\text{diag}}^q \mathbf{A}_i^{-1}$.

The dyadic Green tensor $\overline{\overline{\mathbf{G}}}_{ij}^n$ represents the electromagnetic coupling between dipoles i and j at harmonic n, and includes both the direct (free-space) and reflected (substrate) contributions, such that $\overline{\overline{\mathbf{G}}}_{ij}^n = \overline{\overline{\mathbf{G}}}_{ij}^{0,n} + \overline{\overline{\mathbf{G}}}_{ij}^{r,n}$. The explicit expressions for $\overline{\overline{\mathbf{G}}}_{ij}^{0,n}$ and $\overline{\overline{\mathbf{G}}}_{ij}^{r,n}$ are provided in the Supplementary Section S3. This formulation therefore captures both spatial coupling among all lattice elements through the Green tensors and spectral coupling between harmonics induced by the time modulation of the polarizability.

To solve the system in matrix form, we truncate the harmonic expansion to a finite number N_{h}, corresponding to the range of harmonic orders $n \in [-N_h, N_h]$. All induced dipoles can then be written in a compact matrix formulation by defining, for each harmonic order n, the dipole and electric field vectors as $\mathbf{p}^n = [\,\mathbf{p}_1^n\ \mathbf{p}_2^n\ \cdots\ \mathbf{p}_N^n\,]^{\mathsf{T}}$ and $\mathbf{E}^n = [\,\mathbf{E}_{\text{inc},1}^n\ \mathbf{E}_{\text{inc},2}^n\ \cdots\ \mathbf{E}_{\text{inc},N}^n\,]^{\mathsf{T}}$, where each $\mathbf{p}_i^n$ and $\mathbf{E}_{\text{inc},i}^n$ are 3×1 vectors representing, respectively, the induced dipole moment and the incident electric field at the i-th lattice element for the n-th harmonic.

Stacking all harmonic components, the total dipole and field vectors are expressed as $\mathbf{P} = [\,\mathbf{p}^{-N_h}\ \cdots\ \mathbf{p}^0\ \cdots\ \mathbf{p}^{+N_h}\,]^{\mathsf{T}}$ and $\mathbf{E} = [\,\mathbf{E}^{-N_h}\ \cdots\ \mathbf{E}^0\ \cdots\ \mathbf{E}^{+N_h}\,]^{\mathsf{T}}$, respectively, with $\mathbf{P}, \mathbf{E} \in \mathbb{C}^{(2N_h+1)3N}$. This formulation enables the system of spatially and spectrally coupled dipoles to be written in a single matrix equation $\mathbf{TP} = \mathbf{E}$, where $\mathbf{T}$ contains both the spatial Green-tensor couplings and the cross-harmonic interactions. Collecting the equations of the element for all i and all harmonics, we obtain a single linear system

$$\mathbf{T}\,\mathbf{P} \;=\; \mathbf{E}, \tag{11}$$

where $\mathbf{T}$ is a block–Toeplitz matrix in the harmonic index and block–sparse in the element index. As a matter of fact, $\mathbf{T}$ is responsible for modeling the multiple scattering among the individual dipolar elements through the evaluation of the Green tensor at different Floquet harmonics. We write $\mathbf{T}$ in $(2N_h+1)\times(2N_h+1)$ harmonic blocks $\mathbf{T}_{n,m} \in \mathbb{C}^{3N\times 3N}$, each of which contains $N \times N$ element blocks of size 3×3:

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$$\mathbf{T} = \begin{bmatrix} \mathbf{T}_{-N_h,-N_h} & \cdots & \mathbf{T}_{-N_h,0} & \cdots & \mathbf{T}_{-N_h,+N_h} \\ \vdots & \ddots & \vdots & & \vdots \\ \mathbf{T}_{0,-N_h} & \cdots & \mathbf{T}_{0,0} & \cdots & \mathbf{T}_{0,+N_h} \\ \vdots & & \vdots & \ddots & \vdots \\ \mathbf{T}_{+N_h,-N_h} & \cdots & \mathbf{T}_{+N_h,0} & \cdots & \mathbf{T}_{+N_h,+N_h} \end{bmatrix}.$$

For a fixed harmonic n, the diagonal block $\mathbf{T}_{n,n}$ couples all mesh elements at that same frequency:

$$\big[\mathbf{T}_{n,n}\big]_{i,i} \;=\; \big[\overline{\overline{\boldsymbol{\alpha}}}_j^{n,m}\big]^{-1}, \qquad \big[\mathbf{T}_{n,n}\big]_{i,j} \;=\; -\,\mathbf{A}_i\,\overline{\overline{\mathbf{G}}}_{ij}^{n}\,\mathbf{A}_j^{-1} \quad (i \neq j). \tag{12}$$

Thus, the diagonal blocks 3×3 are the inverse self-polarizabilities at zero shift in harmonic index (no frequency mixing), and the off–diagonals contain the Green–tensor couplings at harmonic n, rotated to the local frames of the i and j elements. The Frequency mixing produced by the time modulation enters as a harmonic convolution on the same element. Hence,

$$\big[\mathbf{T}_{n,m}\big]_{i,i} \;=\; \big[\overline{\overline{\boldsymbol{\alpha}}}_i^{n,m}\big]^{-1}, \qquad \big[\mathbf{T}_{n,m}\big]_{i,j} \;=\; \mathbf{0}_{3\times3} \quad (i \neq j,\; n \neq m). \tag{13}$$

The full description of how to solve the system is available in Supplementary Section S3.Moreover, the validation of the anisotropic and time-modulated DDA is also present in the Supplementary Section S3.6.

Effective polarizability with cross-harmonic coupling

Time modulation transforms each discretized voxel of the multilayer structure into a frequency-mixing dipolar element. After homogenization of the unit cell, shown in Fig. 2(c), the entire multilayer stack behaves as a single effective dipole whose response is described by a harmonic-resolved polarizability tensor. In this scenario, an incident field at harmonic k induces dipole moments not only at k, but also at neighboring harmonics $n = k \pm 1,\ k \pm 2, \ldots$. Consequently, the effective polarizability is no longer a single 3×3 tensor, but a block operator that couples every input harmonic to every output harmonic.

To extract the full tensor, we follow a systematic procedure based on illuminating the homogenized element with one Floquet harmonic at a time. First, the multilayer time-modulated element [Fig. 2(a)] is solved using the time-modulated DDA, and its response is collapsed into a single effective dipole centered in the unit cell. This homogenized element is then used as the building block of the array shown in Fig. 2(c), where the inter-element interaction is computed through the free-space and substrate-reflected Green tensors.

The extraction of the effective polarizability follows directly from the convolutional relation

$$\mathbf{p}_{\mathrm{tot}}^{n} = \sum_{k=-N_h}^{N_h} \overline{\overline{\boldsymbol{\alpha}}}_{\mathrm{eff}}^{n,k}\,\mathbf{E}_{\mathrm{inc}}^{k}, \tag{14}$$

where $\overline{\overline{\boldsymbol{\alpha}}}_{\mathrm{eff}}^{n,k} \in \mathbb{C}^{3\times3}$ maps the incident field at harmonic k to the induced dipole at harmonic n. To determine the entire tensor, we excite the homogenized element with a single nonzero harmonic component of the incident field, i.e., we set $\mathbf{E}_{\mathrm{inc}}^{k} \neq 0$ for one harmonic k while all others are zero. Solving the DDA system yields the induced dipoles $\mathbf{p}^{n}$ for all output harmonics n. Since the system has only one nonzero excitation, each column block of the polarizability tensor is obtained directly as solving the linear equation 14.

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Repeating this procedure for all $k \in [-N_h, N_h]$ reconstructs the full polarizability, which is a block–Toeplitz matrix in the adiabatic regime.
When assembling the homogenized elements into the extended arrays of Fig. 2(c), the polarizability of each meta-atom can be independently tuned by applying a phase shift to the local time-modulation (full details in the Supplementary section S3.4). This phase control effectively rotates the harmonic components of $\overline{\overline{\boldsymbol{\alpha}}}_{\text{eff}}^{n,k}$, enabling us to tailor the coupling between harmonics and individual array elements. As a consequence, both the total emitted power and the induced dipole moment of each element, $\mathbf{p}_i$, can be dynamically controlled through the spatiotemporal modulation profile. This homogenized response is then used to model extended arrays of nanoantennas, as depicted in Fig. 2(c). The explicit computation of array radiation patterns and far-field response is provided in the Supplementary section S3.5.

Quantum Efficiency

For clarity, we explicitly define the quantum efficiency (QE) used throughout this work. The total power dissipated by the quantum emitter (QE) at the nth Floquet harmonic is computed from the harmonic-resolved dipole moment according to Eq. (7),

$$P_{\text{diss}}^{(n)} = \frac{\omega_{31}^{(n)}}{2} \text{Im}\left[\mathbf{p}_{31}^{(n)} \cdot \mathbf{E}_{\text{loc}}^{(n)*}(\mathbf{r}_{\text{QE}})\right], \tag{15}$$

which accounts for the total energy transferred from the emitter to the electromagnetic environment, including both radiative and non-radiative channels. The radiated power at each harmonic is independently obtained from full far-field integration of the scattered fields generated by the dipole moments. The total radiative power is therefore

$$P_{\text{rad}} = \sum_n P_{\text{rad}}^{(n)}, \tag{16}$$

while the total dissipated power is

$$P_{\text{tot}} = \sum_n P_{\text{diss}}^{(n)}. \tag{17}$$

The quantum efficiency is then defined as

$$\text{QE} = \frac{P_{\text{rad}}}{P_{\text{tot}}} = \frac{\Gamma_{\text{rad}}}{\Gamma_{\text{tot}}}, \tag{18}$$

where Γ_{tot} is the total decay rate modified by the environment, as given by Eq. (8).

We begin the results section by optimizing the geometrical parameters of the MIM cavity to tailor its resonance to the zero-phonon line (ZPL) of the diamond NV center, corresponding to a wavelength of 637 nm and an energy of 1.945 eV. At 637 nm, GHz-scale modulation produces picometer-level wavelength shifts that are not resolvable at room temperature due to ZPL phonon broadening, but become resolvable under cryogenic, MHz-linewidth conditions. After identifying the optimal geometry, we investigate the impact of temporal modulation on the radiated power of a single nanoantenna, highlighting the redistribution of energy among Floquet harmonics. We then extend the analysis to one-dimensional arrays of nanoantennas and demonstrate that the modulated structure enables selective energy coupling into a desired harmonic, as well as dynamic beam steering controlled by the individual modulation phases. Finally, we couple the nanoantenna array to the quantum solver to evaluate the population dynamics, emission rates, and the resulting quantum properties of the NV center.

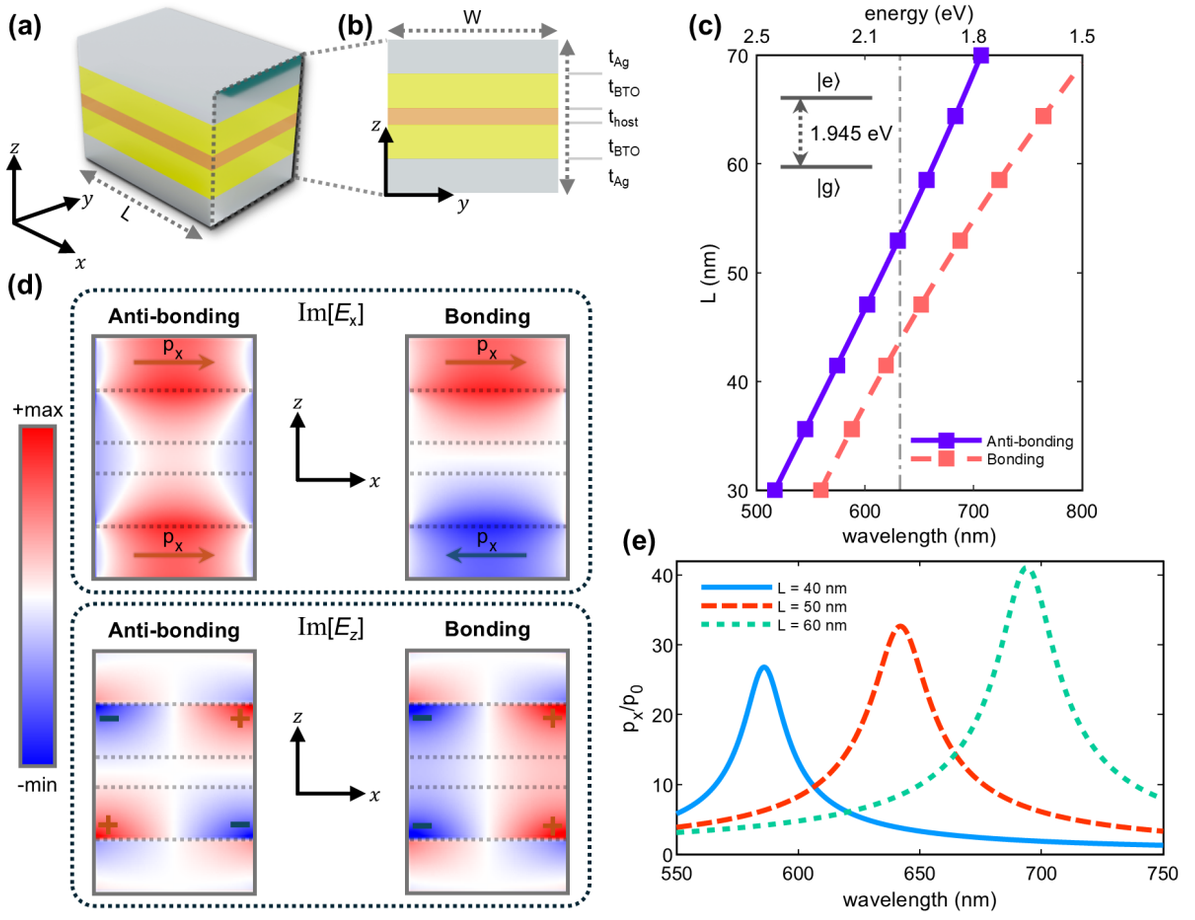


Figure 3: **Mode structure and emitter–cavity interaction in a single MIM meta-atom.** (a) Schematic illustration of the MIM cavity used to manipulate the radiation of a quantum emitter. (b) Side view of the meta-atom alongside its corresponding geometric parameters. (c) Dispersion of the bonding and anti-bonding modes of the MIM cavity with respect to the length of the meta-atom. (d) Electromagnetic field profiles of the two quasinormal modes of the cavity, which explain the symmetry-dependent coupling pathways between the quantum emitter and different modes. (e) Induced electric dipole moment of the nanoantenna (meta-atom), normalized by the fixed excitation dipole moment p_0, for a dipole source placed at the center of the cavity and oriented along x-axis.

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Static cavity response and emitter–mode coupling

In this work, we use an MIM cavity composed of silver cladding layers and a BTO dielectric spacer as the target meta-atom. A thin diamond layer is embedded at the center of the cavity to host an NV-center quantum emitter. The meta-atom schematic is shown in Figs. 3a and 3b, where the geometrical parameters include the cavity length L, width W, and the thicknesses of the constituent layers. The working principles of this cavity and its coupling dynamics to the emitter can be understood in terms of two quasinormal modes, commonly referred to as the anti-bonding and bonding modes. The localized plasmonic modes at the metal–dielectric interfaces hybridize, giving rise to the anti-bonding and bonding modes that exhibit anti-symmetric and symmetric field profiles along the resonator height, respectively, which is a hallmark of coupled plasmonic modes in MIM resonators [64, 65, 66].

As explained, we engineer the cavity resonances to occur around the zero-phonon line (ZPL) of the NV center in diamond, corresponding to a wavelength of 637 nm and an energy of 1.945 eV. The cavity length L is used to fine-tune the spectral response, while the remaining geometrical parameters are set to place the resonances within the desired window, with numerical values given by $W = 10$ nm, $t_{Ag} = 20$ nm, $t_{BTO} = 20$ nm, and $t_{host} = 10$ nm. Quenching in plasmonic nanostructures originates from coupling to high-k, non-radiative modes, which are also the spectral components most affected by nonlocal electronic response [67]. To assess the impact of these effects, we performed an auxiliary planar analysis of the local density of optical states (LDOS) for a dipole above a diamond/BTO/Au stack using the reflected Green function formalism (see Supplementary Section 4.7). By comparing local Drude and hydrodynamic models, we find that nonlocality primarily suppresses high-k evanescent contributions, leading to a reduction of the LDOS only at very short separations. For emitter–interface distances relevant to our configuration (e.g., $h \sim 6\,\mathrm{nm}$ and $d_{\mathrm{BTO}} = 10\,\mathrm{nm}$), the difference between local and nonlocal descriptions remains small, indicating that nonlocal corrections do not significantly affect the radiative efficiency or the overall physical conclusions. Accordingly, nonlocal effects were not included in the DDA calculations.

For the simulation, we have used our in-house anisotropic DDA, with a voxel size of 2nm. The eigenfrequency dispersions of the anti-bonding and bonding modes with respect to L are shown in Fig. 3c. As can be seen, the anti-bonding mode consistently appears at higher energies, while the bonding mode lies at lower frequencies. With meticulous parameter tuning, the two modes can be positioned in the vicinity of the spontaneous emission wavelength of the NV-center quantum emitter in diamond, as indicated by the dashed gray line in Fig. 3c. Both modes exhibit a strong dependence on L, and their highly dispersive nature provides a promising platform for realizing Floquet-driven resonators with deep modulation strengths.

The mode profiles of the anti-bonding and bonding modes in a plane passing through the cavity center are illustrated in Fig. 3d. The effective in-plane dipole moments induced in the metallic plates can be characterized by the electric field component along the x-axis. A symmetric x-directed field corresponds to parallel in-plane electric dipole moments, characteristic of the anti-bonding mode, while an anti-symmetric profile indicates anti-parallel dipole moments associated with the bonding mode. Therefore, owing to the enhanced in-plane dipole moment of the anti-bonding mode, it exhibits strong radiation characteristics. In contrast, absorption is the dominant bonding mode because of its antiparallel in-plane dipole moments. The anti-bonding mode exhibits an odd electric-

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field distribution along the resonator height (x-axis), which enables bright coupling to an x-polarized emitter placed at the cavity center, whereas the bonding mode has even parity and is dark for the same emitter due to the symmetry constraints imposed by the emitter orientation.

This symmetry-dependent coupling behavior is directly reflected in the spectral response of the effective dipole moment displayed in Fig. 3(e). Here, the effective dipole moment refers to the induced dipole of the nanoantenna (meta-atom), calculated using Eq. (12) under excitation by a fixed external dipole source with moment p_0 placed inside the host layer For emitters oriented along the x-axis, the collective dipole response of the MIM cavity exhibits pronounced resonant peaks whose spectral positions strongly depend on the cavity length L. Specifically, increasing L from 40 nm to 60 nm produces a systematic redshift of the antibonding resonance, with peaks centered at wavelengths close to $\lambda \approx 590$ nm, 645 nm, and 705 nm, respectively. At resonance, the effective dipole moment—and consequently the radiated power normalized to free space—can be enhanced by more than one order of magnitude, clearly evidencing strong light–matter coupling mediated by the antibonding plasmonic mode. Among the investigated geometries, the case $L = 50$ nm provides the optimal spectral overlap with the NV-center zero-phonon line at 637 nm, maximizing the dipolar enhancement at the target emission wavelength. When combined with the fast and efficient refractive-index modulation enabled by the BTO layer, this optimized MIM configuration establishes a powerful platform for the active control of the emitter dipole moment and its radiation properties in both space and time via Floquet engineering.

Floquet dynamics and harmonic response of the time-modulated meta-atom

We use BTO, with its strong electro-optic response, to modulate the MIM cavity in time, thereby coupling the diamond NV-center quantum emitter to different Floquet orders (harmonics).In this section, the NV center is modeled as a dipolar quantum emitter with dipolar moment p_0 embedded within the NV-diamond layer. The schematic of the time-varying meta-atom is shown in Fig. 4a, where a sinusoidal voltage is applied between the metallic plates to induce temporal modulation $V(t) = V_0/2\cos(\omega_m t + \varphi_0)$, where V_0 and φ_0 denote, respectively, the modulation amplitude and phase. The cavity length is set to $L = 50$ nm, aligning the anti-bonding resonance with the ZPL of the NV center. A silica base of thickness $t_{base} = 25$ nm is attached to the meta-atom to facilitate its integration into an array or onto a substrate. BTO has a uniaxial crystal structure with its optical axis along the c-axis [40]. Its tetragonal P4mm symmetry allows off-diagonal electro-optic tensor elements, which, under an applied field, introduce cross-coupling terms in the permittivity tensor between the longitudinal and horizontal components (a detailed discussion of the optical properties and electro-optic response of BTO is provided in the Supplementary Section S5). The magnitude of the electric field inside the BTO layer can be estimated from the applied voltage and the electrode spacing. For $V_0 = 2$ V across an effective spacing of $d = 32$ nm, the resulting field is $E \approx 6.25 \times 10^7$ V/m (0.625 MV/cm), which remains below reported dielectric breakdown fields for epitaxial BTO thin films [68]. Hybrid-plasmonic BTO platforms have further demonstrated electro-optic modulation bandwidths exceeding 200 GBd under similar operating conditions [69].Furthermore, the full validation of the anisotropic TM-DDA solver and the convergence tests can be found in Supporting Section 3.6, where good agreement with independent full-wave references

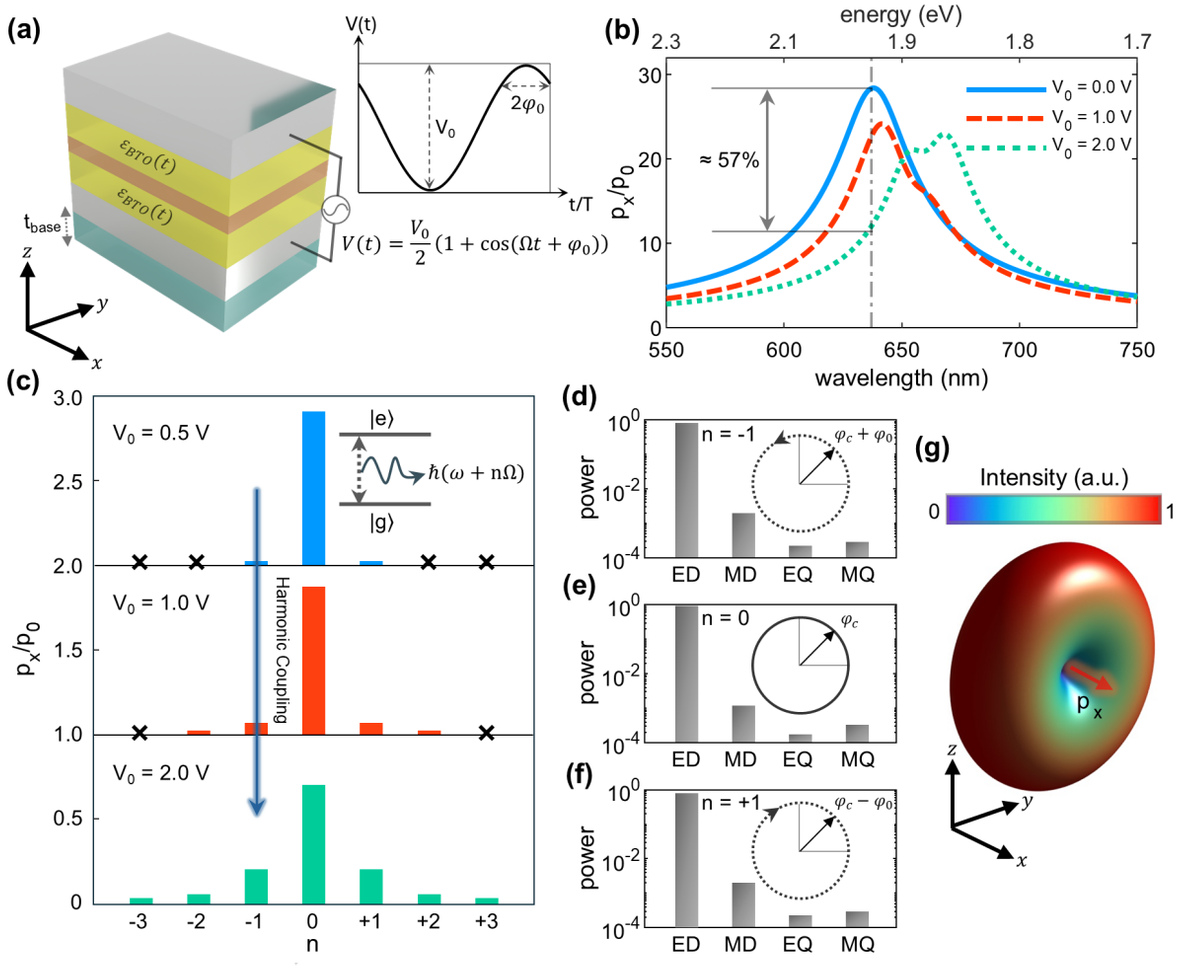


Figure 4: **Floquet response of a time-modulated BTO MIM meta-atom.** (a) Schematic depiction of a time-modulated cavity together with its sinusoidal modulation function, characterized by amplitude V_0, phase φ_0, and angular frequency Ω. (b) Electric dipole moments of the static cavity, calculated with eq. 12, as a function of wavelength for different bias voltages, induced by a diamond NV center oriented along the cavity length (x-axis). (c) Dipole moments of the NV center at $\lambda = 637$ nm for different frequency harmonics at a modulation frequency of $\Omega = 50$ GHz. (d–g) Radiated power of the prominent frequency harmonics (in logarithm scale), normalized to the total emitted power, together with their multipole decomposition. Insets show the dipole-moment phase trajectory as a function of modulation phase, where φ_c denotes the constant phase of the zeroth-order harmonic. (h) Normalized far-field radiation patterns corresponding to the main frequency harmonics.

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is demonstrated.

For this simulation, we employed our in-house time-modulated anisotropic DDA solver. The nanostructures were discretized with a voxel size of 2 nm, which was shown to yield stable spectral features and negligible variation upon further spatial refinement. In the Floquet expansion, harmonic orders from $n = -3$ to $n = +3$ were retained. Higher-order harmonics were systematically tested and found to contribute negligibly. In particular, the power associated with the $n = \pm 3$ harmonics is already more than three orders of magnitude smaller than the fundamental component. This rapid decay in harmonic amplitude confirms convergence with respect to Floquet truncation and ensures that the reported results are not affected by insufficient harmonic resolution.

At higher bias voltages, the bonding mode is no longer dark to an x-oriented quantum emitter, as the symmetry constraints are lifted by the asymmetry introduced through dielectric anisotropy. The induced dipole moments in the cavity for an emitter oriented along x-axis exhibit a clear resonance at $\lambda = 637$ nm, corresponding to the ZPL of the NV center (Fig. 4b). This peak is associated with the anti-bonding mode of the cavity and exhibits a monotonic redshift with increasing bias voltage. Higher bias voltages give rise to a second, weaker peak in the dipole moment spectrum, marking the onset of coupling to the cavity's bonding mode. Together, the coupled plasmonic modes modulate the induced dipole moment by approximately 57% through the electro-optic response of BTO, establishing a robust platform for realizing a Floquet-driven cavity.

The electro-optic response of BTO has been modulated at frequencies in the gigahertz regime, with the speed primarily determined by device constraints rather than the material's intrinsic response [40]. We choose a modulation angular frequency of $\Omega = 50$ GHz, which not only facilitates efficient harmonic coupling but also falls below experimentally demonstrated limits in BTO [69]. Importantly, the proposed framework is general and remains valid for any chosen modulation frequency; varying Ω would modify only the spectral separation between harmonics, without altering the underlying coupling mechanism or the model's applicability. The light–matter interactions between the NV center quantum emitter and the Floquet-driven cavity give rise to dipole moments coupled to different Floquet modes, with oscillation frequencies up- or down-converted by integer multiples of Ω. These dipole moments correspond to radiated photons whose energies are shifted according to the associated Floquet harmonic.

The normalized dipole moments of the quantum emitter in the time-varying cavity at $\lambda = 637$ nm demonstrate that increasing the bias voltage results in stronger coupling to the Floquet harmonics (Fig. 4c). A sinusoidal modulation waveform generates dominant harmonics at $n = \pm 1$, and the nearly symmetric distribution around $n = 0$ arises from the adiabatic nature of the modulation process, given the small modulation frequency ($\Omega/\omega \approx 10^{-4}$). Multipole decomposition of the normalized radiated power for the main Floquet modes confirms that the meta-atom is captured by its electric dipole (ED) response, since the higher-order multipoles, including magnetic dipole (MD), electric quadrupole (EQ), and magnetic quadrupole (MQ), are at least three orders of magnitude weaker and can be safely disregarded (Figs. 4d–g).

This symmetry-dependent coupling behavior is directly reflected in the spectral response of the effective dipole moment displayed in Fig. 4(e). For emitters oriented along the x-axis, the collective dipole response of the MIM cavity exhibits pronounced resonant peaks whose spectral positions strongly depend on the cavity length L. Specifically, increasing L from 40 nm to 60 nm produces a systematic redshift of the antibonding resonance, with peaks centered at wavelengths close to $\lambda \approx 590$ nm, 645 nm, and 705 nm, respectively. At

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resonance, the effective dipole moment—and consequently the radiated power normalized to free space—can be enhanced by more than one order of magnitude, evidencing strong light–matter coupling mediated by the antibonding plasmonic mode.

Among the investigated geometries, the case $L = 50$ nm provides the optimal spectral overlap with the NV-center zero-phonon line at 637 nm, thereby maximizing the dipolar enhancement at the target emission wavelength. The phase insets in Figs. 4(d–g) illustrate how the total phase of each Floquet harmonic arises from two distinct contributions. First, the static MIM cavity exhibits an intrinsic scattering phase φ_c at the carrier frequency, which depends only on the cavity geometry and material parameters and cannot be externally tuned. Second, the applied temporal modulation $V(t) = V_0/2\cos(\Omega t + \varphi_0)$ introduces a controllable phase φ_0, corresponding to the phase of the RF voltage applied to a single meta-atom. Under harmonic modulation, the nth Floquet component inherits a phase shift proportional to the modulation phase, such that $\varphi^{(n)} = \varphi_c + n\,\varphi_0$. Thus, the phase of each harmonic (n) is given by the sum of the intrinsic cavity phase and an integer multiple of the applied modulation phase. The symbols $\varphi_c \pm \varphi_0$ shown in the insets correspond explicitly to the first upper and lower sidebands ($n = \pm 1$).

Importantly, since the modulation phase φ_i of each meta-atom appears explicitly in the definition of φ_0, the phase of every individual Floquet harmonic can be independently programmed by controlling φ_i. As a consequence, not only the spectral distribution of the emitted power but also the spatial phase profile of each harmonic across the array can be dynamically engineered. This enables full spatiotemporal control of the quantum emission, making dynamic beam steering, harmonic-selective wavefront shaping, and directional single-photon routing feasible in time-modulated quantum nanoantenna arrays.

It should be emphasized that the significance of electro-optic modulation does not lie in producing large frequency shifts, but in introducing a programmable synthetic frequency dimension. Even with small harmonic separation, distinct Floquet channels provide dispersionless and electronically reconfigurable control over the complex amplitude of the emitted field. By spatially programming the modulation phase across the metasurface, the phase front of a selected harmonic can be dynamically engineered, enabling harmonic-selective emission routing and active beam steering without mechanical reconfiguration. Because temporal modulation reshapes the electromagnetic Green tensor, it modifies both coherent and dissipative interactions, thereby expanding the accessible control space beyond static nanophotonic platforms.

Although the present implementation employs GHz electro-optic modulation, the formalism is general and directly extendable to all-optical time-varying schemes, such as Kerr or pump-induced refractive-index modulation. Higher modulation frequencies would enhance synthetic-frequency separation and enable broadband frequency conversion, strong nonreciprocal scattering, and access to time-periodic photonic phases, including photonic time crystals [70, 71, 72, 73].

Finally, the resulting impact on the radiation pattern is illustrated in Fig. 4(h), which shows the far-field intensity distribution of the dominant harmonic. The emission exhibits a well-defined directional lobe aligned with the effective electric dipole moment $\mathbf{p}_x$, confirming that the modulated MIM cavity not only enhances the emission strength but also enforces a highly directional radiation profile. When combined with the fast and efficient refractive-index modulation enabled by the BTO layer, this optimized MIM configuration establishes a platform for the active control of the emitter dipole, its emission phase, and its radiation pattern in both space and time via Floquet engineering.

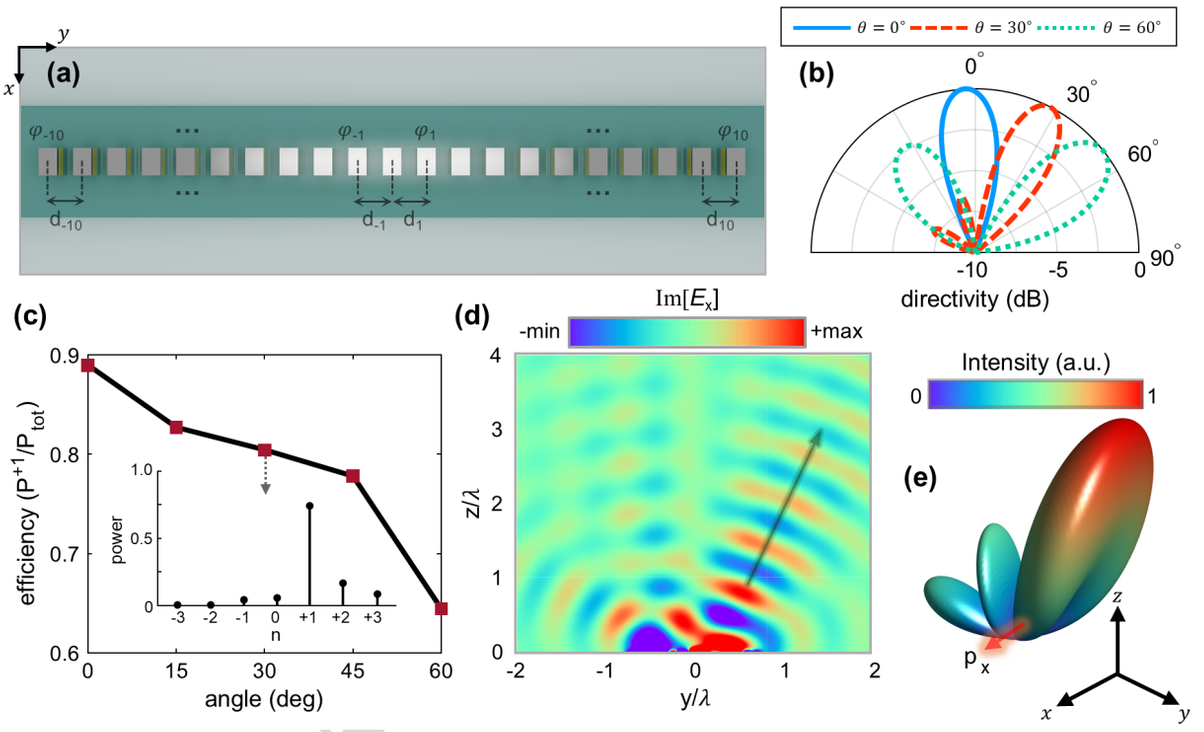


Figure 5: **Floquet-harmonic steering in a time-modulated meta-atom array.** (a) Schematic of the one-dimensional array composed of Floquet-driven BTO meta-atoms, where both the inter-element distances and modulation phases φ_n are optimized to preserve high radiation efficiency. (b) Normalized far-field patterns of the $n = +1$ Floquet harmonic showing programmable beam steering obtained by imposing a phase gradient across the array. (c) Radiation efficiency of the steered harmonic as a function of emission angle, demonstrating stable performance over a wide angular range. (d) Spatial electric-field distribution of the $n = +1$ harmonic, illustrating constructive interference along the steered direction. (e) Three-dimensional radiation pattern of the same harmonic, showing a well-defined main lobe aligned with the programmed steering angle.

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Array-level control of Floquet harmonics in time-varying metasurfaces

The Floquet-driven meta-atom is now incorporated into a one-dimensional array, where both inter-element spacing and modulation phases are optimized to maintain high radiation efficiency while enabling programmable wavefront control. The array schematic is shown in Fig. 5a, where each element is assigned an individual modulation phase φ_n to shape the interference pattern of the radiated Floquet harmonics.

We consider a one-dimensional array of $N = 21$ Floquet-driven nanoantennas, with the NV center located at the central element ($i = 11$), modeled as a fixed dipole source with moment p_0 placed at the central nanoantenna. The ith element is located at coordinate x_i, and the center-to-center spacing between adjacent elements is defined as $d_i = x_{i+1} - x_i$. Each nanoantenna is excited by a time-modulated voltage $V_i(t) = V_{0,i}\cos(\Omega t + \varphi_{0,i})$, where $\varphi_{0,i}$ denotes the applied modulation phase of the ith element. As discussed previously, the phase of the nth Floquet harmonic radiated by element i follows $\varphi_i^{(n)} = \varphi_c + n\,\varphi_{0,i}$, so that spatial variations of $\varphi_{0,i}$ across the array directly determine the phase profile of a selected harmonic through coherent interference. The far-field radiation pattern of that harmonic is determined by the coherent superposition of the contributions from all elements; thus, controlling $\varphi_{0,i}$ enables constructive interference in a desired direction and destructive interference elsewhere, allowing programmable wavefront shaping and beam steering.

The array optimization is performed in two sequential stages to separate geometric design from dynamic beam steering. In the first stage, we jointly optimize the inter-element spacings $\{d_i\}$ and the modulation phases $\{\varphi_{0,i}\}$ for a reference steering angle $\theta_t = 0°$. This stage determines the optimal physical layout of the array together with the corresponding phase distribution that maximizes directional radiation into the $n = +1$ Floquet harmonic.

For a given configuration of spacings and phases, we compute the far-field radiation pattern of the +1 harmonic, $f^{(1)}(\theta, \boldsymbol{\varphi}_0)$, along with the harmonic-resolved radiated powers $P^{(n)}(\boldsymbol{\varphi}_0)$. The optimization is guided by the cost function

$$J \;=\; \left| f^{(1)}(\theta, \boldsymbol{\varphi}_0) - e^{-\left(\frac{\theta-\theta_t}{2\theta_{3\mathrm{dB}}}\right)^2} \right| \;+\; 7\left(1 - \frac{P^{(+1)}}{P_{\mathrm{tot}}}\right), \tag{19}$$

where $\theta_{3\mathrm{dB}} = 5°$. The first term enforces a Gaussian-like main lobe centered at the target angle θ_t, penalizing sidelobes and undesired radiation away from the steering direction. The second term maximizes coupling efficiency into the $n = +1$ Floquet harmonic by penalizing power leakage into other harmonics. The relative weighting between the two terms reflects a deliberate balance between spatial directivity and harmonic purity, which are the two primary performance metrics of interest in this work.

Once the optimal geometry $\{d_i^\star\}$ and reference phase distribution $\{\varphi_{0,i}^\star\}$ for $\theta_t = 0°$ are obtained, the inter-element spacings are fixed. In the second stage, for each new steering angle θ_t, only the modulation phases $\{\varphi_{0,i}\}$ are re-optimized using the same cost function. This separation enables dynamic steering of the +1 harmonic solely via electrical phase control, without altering the physical layout of the nanoantenna array. The complete set of optimized spacings and phase distributions for all considered steering angles is provided in the Supplementary Section S6.

After distance optimization, the retrieved phases are used to steer the emission direction of the array, as shown in Fig. 5(b) for the $n = +1$ harmonic. The normalized far-field

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patterns demonstrate accurate beam steering as the phase gradient is varied, with the main lobe closely following the target angle θ_t. For $\theta_t = 60°$, however, noticeable side lobes arise due to the limited aperture, indicating that the present geometry can reliably steer the beam only up to approximately $\pm 50°$. Increasing the number of meta-atoms would broaden the steering range and enhance directivity, as array gain scales with the number of radiating elements.

A key physical limitation is that the Floquet harmonics have wavelengths $\lambda_n = \frac{2\pi c}{\omega_{31}+n\Omega}$, which differ only weakly because $\omega_{31} \gg \Omega$. As a result, adjacent harmonics are separated by sub-nanometer shifts, making spectral filtering effectively impossible. Any residual radiation from unwanted harmonics would therefore interfere with the far-field pattern of the desired $n = +1$ channel. This constraint motivates concentrating as much energy as possible into a single harmonic. Although an isolated nanoantenna couples only weakly to the $n = +1$ harmonic (Fig. 3(a)), the array benefits from coherent interference between neighboring elements, enabling higher power concentration into this Floquet channel. This is confirmed in Fig. 5(c), where the coupling efficiency $\eta_{+1} = \frac{P^{(+1)}}{P_{\text{tot}}}$ decreases from $\sim 90\%$ at $\theta_t = 0°$ to $\sim 65\%$ at $\theta_t = 60°$. The inset of Fig. 5(c) shows that the remaining harmonics carry negligible energy and radiate nearly isotropically, contributing very little to the directive emission observed in Fig. 5(b). The spatial field map in Fig. 5(d) further highlights constructive interference along the programmed steering direction and suppressed radiation elsewhere, demonstrating the effectiveness of combining optimized spacing with phase control to achieve directive emission in a single Floquet harmonic.

The spatial field profile of the $n = +1$ Floquet harmonic is shown in Fig. 5d, where a pronounced forward-directed wavefront emerges from the optimized array. The normalized electric field along the emitter axis exhibits strong constructive interference in the steered direction and suppression elsewhere, demonstrating the synergy between optimized inter-element spacing and tailored modulation phases. The corresponding three-dimensional radiation pattern in Fig. 5e further confirms a well-defined main lobe aligned with the target emission angle, with minimal residual radiation in undesired directions. Nonetheless, for large steering angles, as shown in Fig. 5b, considerable radiation sidelobes start to appear. Their origin can be traced back to the spatial incoherence of the QE, along with the finite size of the array, which limits the effective aperture and thus the achievable spatial resolution. This behavior is different in nature from aperiodic metasurface platforms, where a spatially coherent excitation (e.g., a plane wave) directly drives all array elements, enabling a more independent control of their relative phases and amplitudes and therefore allowing for stronger sidelobe suppression [74, 75, 76].

Together, these results show that arrays of time-modulated BTO meta-atoms can be engineered to redistribute optical power into specific Floquet harmonics with high angular precision, enabling electronically programmable directive emission. More broadly, the combined control over element spacing, modulation phase, and harmonic selection establishes time-modulated BTO meta-atoms as a platform for dynamic beamforming at the quantum scale.

It is important to highlight that, while the present work demonstrates the physical capability of harmonic-selective beam steering through independently programmable phase channels, practical implementation at tens of GHz requires careful RF engineering. Challenges include impedance matching, parasitic capacitance, electrode cross-talk, and thermal management in densely packed nanoantenna arrays. In realistic device architectures, independent phase control may be implemented using grouped or row-column addressing schemes, shared-bus electrode networks, or pixelated modulation domains rather

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than fully independent nanoscale wiring. Electrically tunable metasurfaces with programmable spatial phase profiles have been experimentally demonstrated across multiple active-material platforms, including conducting-oxide-gated metasurfaces [77], graphene-metal reconfigurable metasurfaces [78], electrically contacted nanoantenna matrices [79], and, more recently, space-time metasurfaces operating at optical frequencies [80]. The broader framework of spatiotemporal light control with active metasurfaces has also been comprehensively reviewed in [81]. These works collectively establish that electrically addressable metasurfaces with programmable phase control are experimentally viable, although scaling toward deeply subwavelength optical arrays with fully independent GHz-per-element wiring remains an important engineering frontier.

Having established the beam-steering behavior of the modulated array, we now turn to the quantum aspect of the problem. Specifically, we investigate how the optimized nanoantenna array modifies the local photonic density of states experienced by an NV center placed at the central meta-atom, and how this, in turn, governs the time evolution of the quantum-emitter density matrix and its harmonic-resolved emission dynamics.

Floquet–Lindblad Dynamics of an QE Coupled to Time-Modulated Nanoantennas

We now analyze the quantum-optical response of a nitrogen–vacancy (NV) center interacting with a time-modulated BTO nanoantenna platform. The emitter is modeled as a three-level system with decay rates $\gamma_{21} = 2\pi/80\,\text{ns}$, $\gamma_{23} = 2\pi/60\,\text{ps}$, and $\gamma_{31} = 2\pi/13\,\text{ns}$, with optical transitions at $\lambda_{12} = 532$ nm and $\lambda_{13} = 637$ nm [82, 83, 84]. The NV center is initially prepared in the ground state ($\rho_{11} = 1$), positioned at the central nano-antenna of the array and is coherently driven through a Hamiltonian coupling term with Rabi frequency $\Omega_{12} = 10$ MHz, and its dynamics are obtained from the Floquet–GKSL formalism described previously. Figures 6 (a-f) show the dissipative and radiative behavior of the emitter. Panels (a–c) present the total dissipated power $P_d^n(t)/P_{\text{free}}$, where $P_d^n(t)$ is the harmonic-resolved power dissipated by the emitter and P_{free} is the corresponding free-space value. Panels (d–f) show the radiated fraction $P_{\text{rad}}^n(t)/P_{\text{free}}$. Additional harmonics ($-2 \leq n \leq 2$) are provided in the Supplementary Fig. S5. Moreover, to validate our dynamics approach, we also show the trace conservation and positivity in Supplementary Fig. S6.

As shown in Figs. 6(a–c), the harmonic-resolved dissipated power $P_d^n(t)$ reveals how time modulation reshapes the dissipative channels available to the quantum emitter. For a single, isolated metal–insulator–metal (MIM) nanoantenna, the local photonic density of states is governed by the bonding resonance intrinsic to the cavity geometry. This results in a strongly localized enhancement of the spontaneous emission rate in the fundamental Floquet channel ($n = 0$), reaching peak values up to 5.6×10^3 relative to free space. In this regime, the emitter couples predominantly to a single resonant decay pathway, with only minor spectral leakage into neighboring harmonics. Although weak, decay channels associated with the $n = \pm 1$ harmonics are present and arise directly from the imposed temporal modulation, a feature that is absent when the emitter is modeled as a classical dipole.

When the emitter is embedded within the optimized 21-element array, the dominant dissipative pathway shifts to the $n = +1$ Floquet harmonic. The array is specifically engineered to transfer radiated power into this harmonic through controlled near-field coupling and interference among adjacent nanoantennas. As a consequence, the electro-

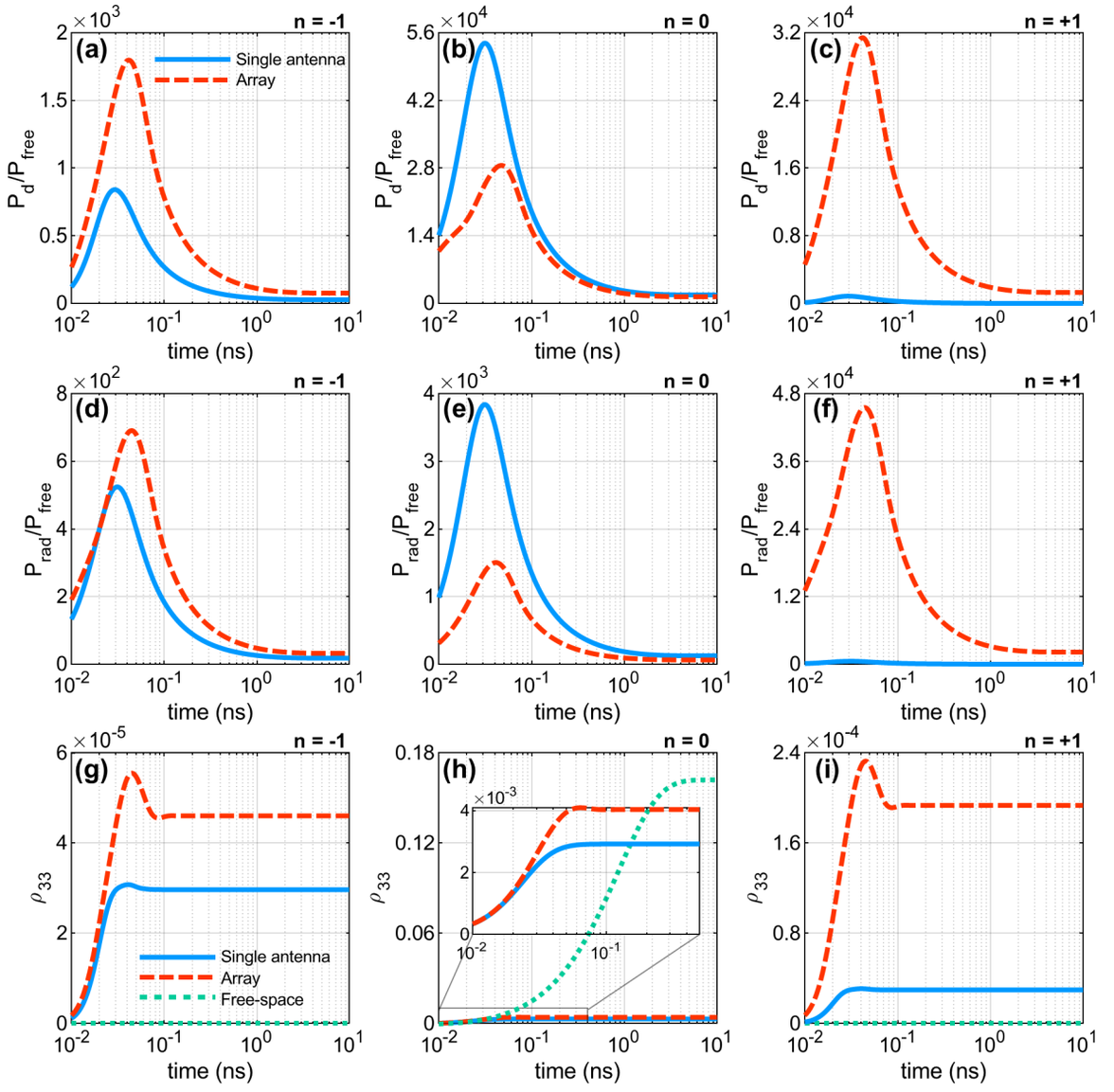


Figure 6: (a–c) show the harmonic-resolved dissipated power $P_d^n(t)/P_{\text{free}}$ for $n = -1, 0, +1$, for a single nanoantenna (blue) and with that 21-element array (red). (d–f) show the radiated power $P_{\text{rad}}^n(t)/P_{\text{free}}$for a single nanoantenna (blue) and with that 21-element array (red). Panels (g–i) display the excited-state population $\rho_{33}^n(t)$ for each harmonic. Harmonics from $n = -2$ to $n = +2$ and additional population dynamics are provided in the Supplementary Fig. S5.

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magnetic fields acting on the quantum emitter enhance the local density of states associated with the $n = +1$ channel, thereby increasing emission at this frequency. Within the Lindblad formalism, this behavior corresponds to an enhancement of the harmonic-dependent decay rate for the $3 \to 1$ transition, Γ_{31}^{n} (see Eq. (7)), and thus to a higher effective emission rate. In contrast, the dissipated power in free space evolves more slowly, since the emitter lifetime is not modified in the absence of the MIM environment. This leads to a reduction of the ratio $P_d^n(t)/P_{\text{free}}(t)$ at longer times for both the single-antenna and array configurations. Physically, this behavior reflects the Purcell enhancement, which shortens the excited-state lifetime by opening additional decay channels. Such lifetime reduction directly translates into higher emission rates, which can be exploited to enhance the information-transmission capacity of photon-based quantum sources.

When examining the fraction of dissipated power that is effectively converted into radiated power, $P_{\text{rad}}^n(t)$, as shown in Figs. 6(d–f), distinct behaviors emerge for the single nanoantenna and the optimized array. For an isolated MIM nanoantenna, the radiated power is predominantly emitted in the fundamental Floquet channel ($n = 0$), in agreement with the classical response shown in Fig. 4(c). The power coupled to the first-order sidebands ($n = \pm 1$) is approximately one order of magnitude smaller, corroborating earlier electromagnetic calculations and confirming that temporal modulation alone produces only weak frequency conversion in the absence of collective effects. In contrast, when the quantum emitter is embedded within the optimized array, the dominant radiative channel shifts to the $n = +1$ harmonic, consistent with the array design strategy that aims to concentrate emission into this sideband through engineered near-field coupling and phase-controlled interference. In an idealized scenario based solely on classical field optimization, the coupling efficiency to the $n = +1$ harmonic would approach $\sim 90\%$. However, this limit is not reached when the full quantum dynamics are taken into account.

The Lindblad–Maxwell framework reveals that the MIM environment also converts a non-negligible fraction of the dissipated power associated with neighboring harmonics into radiated power, thereby reducing the net spectral selectivity. As a result, the effective coupling efficiency to the $n = +1$ harmonic is reduced to approximately 70% when quantum dissipation pathways are properly included. Despite this redistribution, the overall external quantum efficiency is substantially enhanced in the array configuration. Defining the external quantum efficiency as $\eta_{\text{ext}} = \sum_n P_{\text{rad}}^n / \sum_n P_d^n$, we find that η_{ext} increases from roughly 4% for a single nanoantenna to nearly 75% for the optimized array. This marked improvement underscores the critical role of engineered spatial interference and collective electromagnetic interactions in enabling efficient extraction of quantum emission into free space.

Figures 6(g–i) present the temporal evolution of the excited-state population $\rho_{33}^n(t)$ for the three dominant Floquet harmonics. Although the instantaneous population stored in level $|3\rangle$ is reduced when the emitter is placed inside the MIM environment, the optical flux through this level is substantially enhanced due to the increased local density of optical states. In other words, while fewer electrons accumulate in ρ_{33} at any given time, the rate at which electrons are promoted to and decay from this state is significantly higher. As a consequence, the total dissipated power associated with the $3 \to 1$ transition increases, consistent with the large Purcell enhancements previously observed in Figs. 6(a–c). Temporal modulation further modifies this behavior by altering the flux of electrons arriving at and leaving the excited state. The periodic modulation redistributes the decay dynamics across Floquet harmonics, which is explicitly captured by

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the harmonic-resolved populations $\rho_{33}^{n}(t)$. These harmonic populations directly reflect the time-dependent balance between excitation and decay processes and are intimately connected to the dissipated power in each channel. In free space, the long intrinsic lifetime $\tau_0 \approx 13\,\mathrm{ns}$ leads to a slow accumulation of population in ρ_{33} and a correspondingly gradual increase of the downward optical flux. Under Purcell-enhanced conditions, population dynamics are strongly accelerated: the population rises and decays rapidly as additional decay channels open.

Numerical fits to the population decay yield effective lifetimes of approximately 85 ps for a single nanoantenna and 40 ps for the optimized array, corresponding to a nearly two-order-of-magnitude reduction relative to free space. The emergence of sizable $\rho_{33}^{\pm 1}(t)$ components in both the single-antenna and array configurations highlights the role of Floquet driving in redistributing electronic flux among harmonics. These modulation-induced population channels, which directly contribute to harmonic-dependent dissipation, cannot be captured within classical dipole descriptions. Observing the full set of harmonic-resolved population dynamics is therefore essential for understanding and engineering the emitter's quantum decay pathways. In this sense, time modulation provides a powerful and versatile tool for tailoring the quantum emission process, enabling controlled redistribution of electronic flux and dissipation across selected Floquet channels.

Discussion

In conclusion, we have developed a self-consistent integration of quantum open-system dynamics with a full-wave electromagnetic solver in the presence of time-modulated, anisotropic nanostructures, and demonstrated that this framework enables active beam steering and harmonic routing of quantum emission. We have shown that Floquet-driven time modulation of meta-atoms provides programmable control of quantum light, allowing simultaneous manipulation of emission frequency components, radiation efficiency, and angular distribution. In particular, we designed MIM meta-atoms based on BTO multilayer nanoantennas and periodically modulated them to transfer energy into a selected Floquet harmonic while imprinting a dispersionless, electronically programmable phase onto that harmonic. This mechanism enables phase control that is not limited by static geometrical phase engineering or mechanically reconfigurable architectures.

By optimizing the array configuration, this programmable phase control allows shaping and steering of quantum-emitter radiation without relying on static phase masks or mechanical motion. Building on this platform, we extended a general modeling framework for quantum emitters embedded in time-modulated nanophotonic environments, combining harmonic-resolved GKSL dynamics with time-modulated Maxwell electrodynamics solved via an anisotropic DDA. This Floquet–Lindblad–DDA formalism captures the quantum–classical feedback induced by temporal modulation and enables quantitative evaluation of population dynamics, harmonic-resolved dissipated and radiated power, Purcell enhancement, and external quantum efficiency.

Applied to a time-modulated meta-atom platform based on BTO multilayer nanoantennas, the framework reveals how strong electro-optic modulation can be exploited to engineer the local density of optical states simultaneously in space, time, and frequency. We demonstrate that collective interference in an optimized array produces strong lifetime reduction, Purcell factors exceeding 5×10^3, and efficient spectral routing of quantum emission, converting more than 75% of the total dissipated power into a single Floquet

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sideband. This redistribution directly increases the external quantum efficiency, which rises from a few percent for a single nanoantenna to values approaching 75% for the optimized array, highlighting the critical role of engineered interference in enabling efficient free-space extraction of quantum emission. Moreover, by independently controlling the temporal modulation phase of each meta-atom, the array enables electronic beam steering of the selected Floquet harmonic over a wide angular range, from $-50°$ to $50°$.
This capability demonstrates that temporal modulation provides simultaneous control over emission frequency, radiation efficiency, and far-field directionality, without mechanical motion or static phase masks. Beyond the specific platform considered here, the proposed formalism is general and applicable to arbitrary geometries, materials, and modulation schemes, offering a powerful tool for the design and optimization of time-varying quantum photonic systems in regimes where classical or static descriptions are inadequate. These results offer a pathway toward time-modulated nanophotonics as a viable approach to programmable quantum-light sources, reconfigurable quantum metasurfaces, and dynamically tunable architectures for quantum communication, sensing, and information processing.

Methods

Methods

Finite-element simulations of the static nanoantenna

To validate the electromagnetic response of the proposed nanoantenna geometry, full-wave simulations of a single metal–insulator–metal (MIM) cavity were performed using the finite-element solver COMSOL Multiphysics. The structure consisted of two silver layers separated by a BTO dielectric spacer and embedded in a host diamond layer containing the NV center emitter. The metal permittivity was modeled using experimentally reported dispersive optical constants for silver, while the dielectric response of BTO was modeled as an anisotropic medium with refractive indices consistent with experimental measurements (see Supplementary Section S4). The computational domain was surrounded by perfectly matched layers (PMLs) to emulate open-boundary radiation conditions. A dipole source representing the NV center was placed at the center of the diamond host layer. The dipole was oriented along the cavity axis (x-direction), thereby maximizing coupling to the cavity's antibonding plasmonic mode. The dipole radiation and cavity response were evaluated using the frequency domain solver over a spectral range around the NV zero-phonon line (ZPL) at $\lambda = 637$ nm. From the computed electromagnetic fields, the nanoantenna's induced dipole moment and the corresponding radiated power were extracted. Additionally, the eigenfrequency solver was used to determine the resonance frequencies of the antibonding and bonding modes as a function of geometric parameters. These simulations were used to identify the optimal cavity length L that aligns the antibonding resonance with the NV ZPL, and to validate the resonance behavior predicted by the discrete dipole framework (see Supplementary Fig. S3).

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Time-modulated single nanoantenna simulations

The time-modulated response of the nanoantenna was computed using an in-house anisotropic time-modulated discrete dipole approximation (TM-DDA) solver. The nanoantenna structure was discretized into cubic voxels with side length 2 nm, resulting in a lattice of dipolar elements whose permittivity tensor varies periodically in time due to the electro-optic response of BTO. Temporal modulation was introduced through the Pockels effect in the BTO layer. The permittivity tensor of BTO was updated according to the applied voltage $V(t) = \frac{V_0}{2}\cos(\Omega t + \varphi_0)$, which induces a time-dependent refractive index variation and therefore a time-dependent polarizability tensor for each discretized element. The electro-optic coefficients used in the simulations correspond to experimentally reported clamped values for BTO (see Supplementary Table S1). Because the system is periodically modulated, the electromagnetic fields were expanded in a Floquet harmonic basis following the formulation described in Supplementary Section S1, where the time-dependent permittivity is written as a Fourier expansion (see Supplementary Eq. S1). Harmonics from $n = -3$ to $n = +3$ were retained in the calculations. Convergence tests confirmed that higher-order harmonics contributed negligibly to the total emitted power. In this stage of the study, the NV center was approximated as a classical electric dipole emitter with a fixed dipole moment p_0 located at the cavity center. This approximation is valid for evaluating the nanoantenna's electromagnetic response and identifying the harmonic redistribution induced by temporal modulation.

Time-modulated nanoantenna array simulations

After characterizing the single meta-atom, the system was extended to a one-dimensional array of time-modulated nanoantennas. The array consisted of $N = 21$ elements arranged along the x-direction. Each element was modeled using the effective harmonic polarizability extracted from the homogenized unit cell. The electromagnetic coupling between array elements was computed using the dyadic Green tensor formalism, which accounts for both free-space radiation and substrate-reflected fields (see Supplementary Section S1.2). The induced dipole moment of each element was obtained by solving the coupled dipole equations including cross-harmonic interactions generated by the temporal modulation (see Supplementary Eq. S8). Independent modulation phases were assigned to each element through $V_i(t) = V_{0,i}\cos(\Omega t + \varphi_{0,i})$, which allows spatial control of the phase of the Floquet harmonics radiated by the array. The phase distribution $\{\varphi_{0,i}\}$ was optimized to maximize radiation into the $n = +1$ Floquet harmonic while suppressing emission into other channels. The far-field radiation pattern of each harmonic was obtained from coherent superposition of the radiated fields from all dipoles. This approach enabled evaluation of beam steering and harmonic-selective emission resulting from the spatiotemporal modulation of the array.

Floquet–Lindblad quantum-emitter dynamics

To model the quantum emitter dynamics, the NV center was described using a three-level open quantum system governed by the Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation (see Supplementary Eq. S13). The Hamiltonian includes the dipole interaction between the emitter and the local electromagnetic field produced by the nanoantenna structure. Because the photonic environment is periodically modulated, the density matrix of the emitter was expanded in a Floquet harmonic basis as described

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in Supplementary Section S2. The resulting system of equations couples the harmonic components of the density matrix through the time-dependent electromagnetic fields computed by the TM-DDA solver. The NV center was initialized in the ground state and driven by an external optical pump corresponding to the 532 nm transition. The optical decay channel corresponding to the ZPL transition at 637 nm was coupled to the time-modulated electromagnetic environment provided by the nanoantenna array. At each time step, the harmonic dipole moment of the emitter was computed from the density matrix and used as a source in the electromagnetic solver. The resulting local fields modify the radiative decay rates through the local density of optical states. This procedure establishes a self-consistent quantum–classical feedback loop between the GKSL quantum solver and the time-modulated Maxwell equations. From the harmonic dipole moments and electromagnetic fields, the dissipated power, radiated power, Purcell enhancement, and external quantum efficiency were extracted for each Floquet harmonic.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to their large size and because they are generated using in-house simulation code, but are available from the corresponding author on reasonable request.

Code availability

The codes used to perform the simulations and generate the results of this study are available from the corresponding author on reasonable request.

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Author Contributions Statement

A. F. M. and M. M. S. have equally contributed to the manuscript. A. F. M. performed simulations, obtained the results, discussed the results, analyzed the data, and helped write the paper. M. M. S. conducted simulations, prepared all figures, discussed the results, and assisted in writing the paper. H. M. supervised the work, helped review and discuss all the results. All the authors read and approved the final manuscript.

Competing interests

The authors declare no competing financial or non-financial interests.

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